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di Torino**
Dipartimento
di Scienza Applicata
e Tecnologia

Mechanical Engineering *Technology of Metallic Materials TMM*

Academic Year 2025-2026

Introduction to the course



Lecture 1

Lectures/Exercises/Laboratories: [Giovanni Maizza \(giovanni.maizza@polito.it\)](mailto:giovanni.maizza@polito.it)

Tutorials: [Sergio Perero \(sergio.perero@polito.it\)](mailto:sergio.perero@polito.it)

OUTLINE: TMM Course

- Structure of the course:
42h LECTURES + 8h EXERCISE/TRAINING CLASSES + 6h Tutorials
- Exam Assessment and grading
WRITTEN TEST (multiple choice) + ORAL TEST
- Educational material
- Pre-requisites (STM)
- TMM Objectives, expected learning outcomes
- Assignments, intermediate checks (Tutorials)

TMM Lecture n.1:

- Importance of alloys in mechanical engineering
- Metallic bonding and properties, Condon-Morse model
- Crystal packing and interstitial sites
- Crystal anisotropy and microstructure anisotropy (texture)



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GRADING AND TMM EXAM RULES

RULE upon attending classes: every 20-30 min, the teacher will break the lecture for few minutes to allow Student's questions – NO INTERRUPTIONS ALLOWED

- No past-year TMM videolectures are available, slightly changed syllabus

Extra bonus mark to be added to final TMM exam mark: write down a TMM report on a topic selected by the Student from a complete list of subjects uploaded by the teacher

- 6 Tutorials by prof. Perero in parallel with lectures (+ exercise assessment)

Admission, Exam and Grading

20 min Written «multiple-choice» test (WT) + obligatory Oral test (OT)

- WT & OT inspired by 200 questions collected at the end of lectures
- **Mandatory WT Questions** : phase diagrams, CCT/TTT diagrams
- Excellent student path: n.4 individual in-depth study Assignment (EDA)

DSA delivery: EDA deadline/delivery (by email to giovanni.maizza@polito.it): before next Tutorial Delivery of Technical report: Dec. 20th, 2025

TMM Exam mark = WT + OT + (0 – 3) Written Report

EDA WILL BE USED FOR INSPIRING EXTRA QUESTIONS IN THE ORAL TEST

EDUCATIONAL MATERIAL

- - Lectures' handouts: pre-uploaded on Polito-course Materials portal
- - Aid notes (*upon student's request by email*)

Suggested textbooks:

- P.L. Mangonon, The principles of materials selection for engineering design, Prentice Hall, 1999.
- W.F. Smith, Structure and properties of engineering alloys, 2nd ed. Mc Graw Hill, 1993.
- Meyers-Chawla - Mechanical Behavior of Materials 2nd ed. 2009
- Z. Marchiniak et. al.- Mechanics of Sheet Metal Forming, 2002
- W.F. Smith, Materials Science and Technology, MacGrawHill, 5th International Ed., 2007.
- **It is not recommended to study TMM contents on the Callister's, Materials Science textbook**



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STM Pre-requisites 1/2

- Electronic structure of the elements
- Periodic Table and their peculiar periodic properties
- Electronegativity concept and its role in the interatomic bonding
- The Condon-Morse model: Young modulus and thermal expansion
- Packing factor in crystalline structures (bcc, fcc, hcp),
- Close packed planes and directions
- Miller indices for planes and direction (MMT)
- Polymorphism; anisotropy
- Lattice defects (interstitials, vacancies, dislocations, grain boundaries, stacking faults)
- Conversion of weight-to-atomic concentrations
- Activated processes; Arrhenius law, Diffusion (MMT)
- Engineering and true stress-strain curves in tensile test
- Concept of normal stress and shear stress
- Yield stress, Young modulus and tensile strength; Hooke's law



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STM Pre-requisites 2/2

- Hardness tests, Charpy test, Fatigue test, Creep test
- Fracture mechanisms: ductile, brittle (cup-cone fracture); transgranular and intergranular fracture (cleavage), embrittlement
- Fatigue and fatigue resistance; crack propagation; critical factors
- Fe-C metastable phase diagram in the steel range: application of lever rule to the determination of the weight fraction of phases and %C of the transforming phase on heating and cooling

Because Heat Transfer course is not provided before TMM course anymore some fundamentals on conduction & convection heat transfer will be provided by written notes for self-learning



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TMM Topics

Lattice, Miller indices exercises

Condon-Morse model: Young modulus and thermal expansion

Diffusion and thermally activated phenomena + exercises

Corrosion

Stainless Steels and applications

Aluminum alloys and applications

Fe-Fe₃C metastable phase diagram + exercises

Metals, electronegativity, polymorphism, anisotropy

Microplasticity and plasticity

Exercises on alloy strengthening and plasticity

Shaping by casting and solidification

Shaping by plastic deformation

Steels and cast iron (ferrous alloys), tool steels + nomenclature

Surface Heat treatment of steels

Non-ferrous alloys (Mg-alloys, Ti-alloys, Cu-alloys)

Instrumented Indentation test exercise

Selection of alloys

n.4 Tutoring + Assignments (1:30h each)



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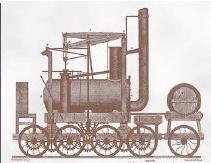
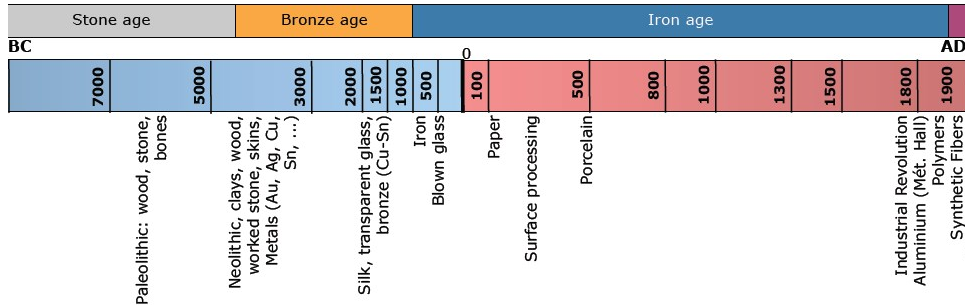
TMM Objectives and Expected Outcome

At the end of TMM course the student must be able to:

- Learn metallurgical fundamentals and cross-link phenomena
- Practicing with the fundamentals of equilibrium Fe-C phase diagrams to equilibrium heat treatment of steels
- Learn and practicing with the fundamentals of TTT and CCT diagrams to non-equilibrium heat treatment of steels
- Correlate the microstructure, mechanical properties and manufacturing process conditions and apply them to:
 - mechanical design of components and structures
 - Select the optimal shaping process for specific eng.ng
- Select optimal metals and alloys for mechanical applications while meeting safety, durability and sustainability requirements



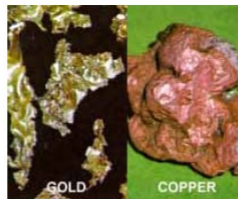
Historical evolution of materials



Historical evolution of Metals



300,000 BC flint, can be easily shaped for tools, beginning of ceramics



5,500 BC native gold and copper as found on surface of earth used for tools and weapons



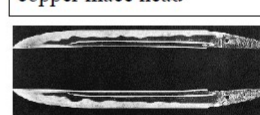
4,000 BC, melting and casting of metals, earliest known cast metal object, copper mace head



3,500 Egyptian tomb painting recording metal working



~ 3,000 BC, bronze, the first alloy, metallurgy



Damascus sword, Fe:C alloys, steel, for blades, guns, canons, tanks, airplanes, depleted U ammunition to destroy tank armor,

METALS of the 2000 age



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High-performance Steels: Myths & Legends

1000 BC – 650 AD, India – wootz iron → Damascus steels

small pieces of *wrought iron* together with some *wood chips and leaves* in small clay crucible which were sealed with a clay lid and then heated in air blast *enhanced fires*, this enabled (less than 2%C) to distribute uniformly, resulting in a **sword** of superior *robustness, toughness and sharp edge*

From von Stahel and Eysen, Nurmberg, 1532, by German Alchemist

Take varnish, dragon's blood, horn scraping, half as much salt, made from earthworms, radish juice, tallow and vervain and quench therein. It is also very advantageous in hardening if a piece that is to be hardened is first thoroughly cleaned and well polished. ...

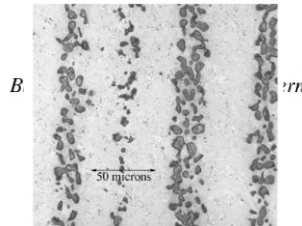
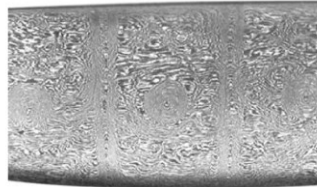
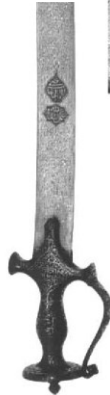
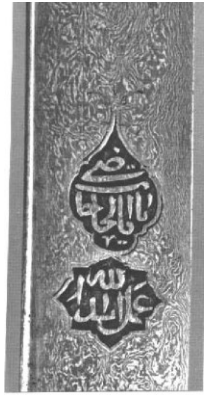
If fault should be found with some of the recipes, pray to not reject the whole book. Perhaps the fault lies in the user himself, because he did not follow the instructions correctly. All arts requite practice and long experience, and their mastery is only gradually acquired.

Legend: **medieval German weapon smith Wieland's recipe:** *fiel a forged piece of iron into coarse powder, fed to your chicken, separate filing from the chickens faces by a magnet, after 7 passes through the chicken bodies, you have the material for swords of superior strength and hardness.*



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The Mistry of Damascus Swords



Lecture 1a

The inside of a Damascus blade, magnified 370times, showing the carbide particles aligned as rows.

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The following paper, The Mystery of the Damascus Sword by John Verhoeven and Alfred Pendray appeared in Muse, Volume 2, Number 2, pp. 35-43, April 1998.

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TMM Lecture n.1 Fundamentals



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Materials Science and Engineering

Bernhardt Wuensch at MIT: **materials science** is primarily about “*relation between the structure of matter and its properties*”

materials engineering is primarily about “*modification of properties and performance during and after processing, and with manufacture*”,



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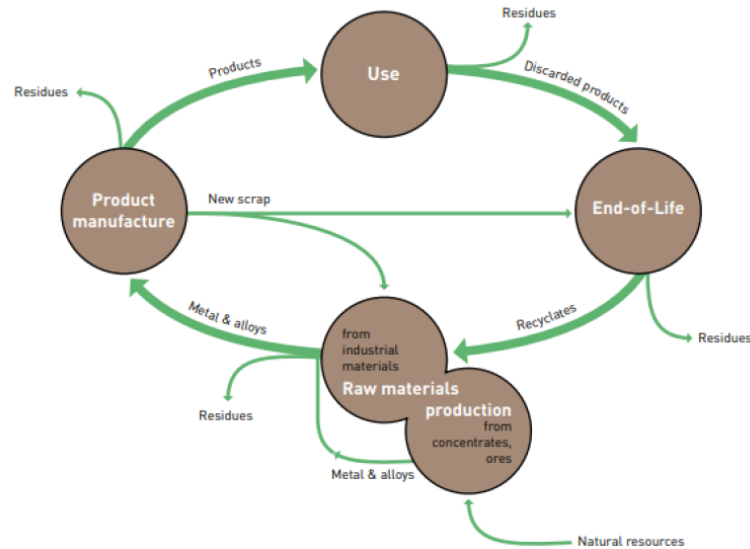
Engineering and Technology TRENDS for the Future

- **Low price + higher value** of the component
- **Lower cost** under **LCA constraints** of the product
 - Multimaterials systems
- **Higher quality**, **Safety**, **superior Performances**
 - **Higher resistance** combined with **higher toughness**
 - Higher **corrosion resistance**, **wear resistance** and higher **resistance to oxidation** to high temperatures
 - **Enhanced weldability**, **formability** and **machinability**
- **Environmentally sustainable** materials
- **Design for disassembly**



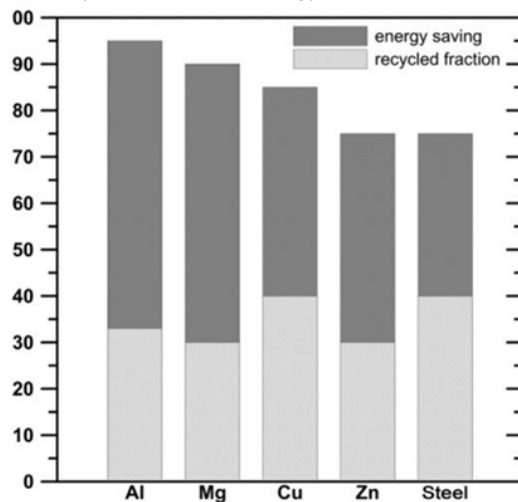
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ISTANABILITY OF MATERIALS AND PROCESSES



FRACTION OF RECYCLED ALLOYS

The total energy use is the most meaningful indicator, due to the relative ease of assessing this, and the fact that CO₂ production is the major concern today, and this is closely related to the energy.



11/10/2025

Source 2016

Compared to other **high-volume materials**, such as copper (Cu), zinc (Zn), magnesium (Mg), and steel, Al production has one of the widest energy differences between the primary and secondary routes, but not the **main recycled fraction**, that is the share of secondary production with respect to the total one (Figure 1). The **recycled Al fraction is about 35%**, which is close to the values of recycled Mg and Zn (~30%). Nowadays, **copper and steel remain the materials with the highest impact in terms of recycled amounts (~40%)**.

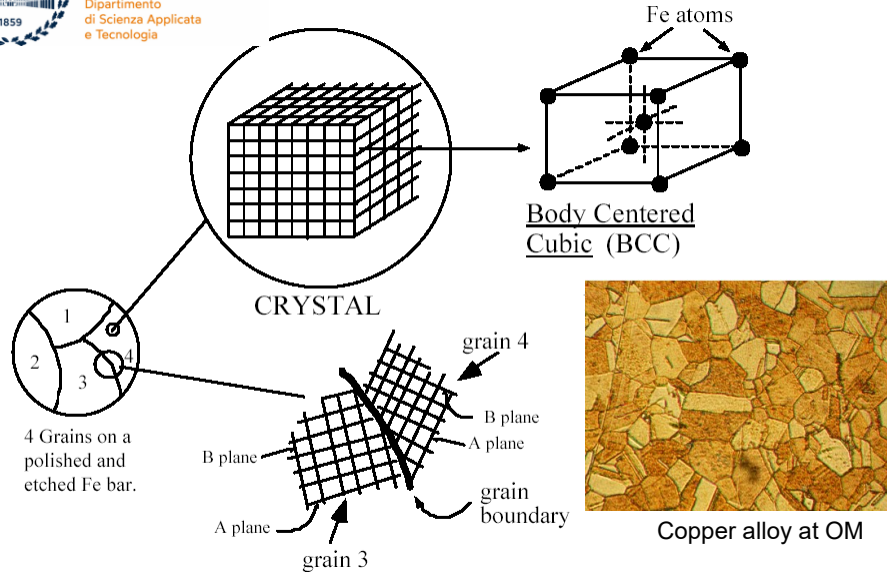


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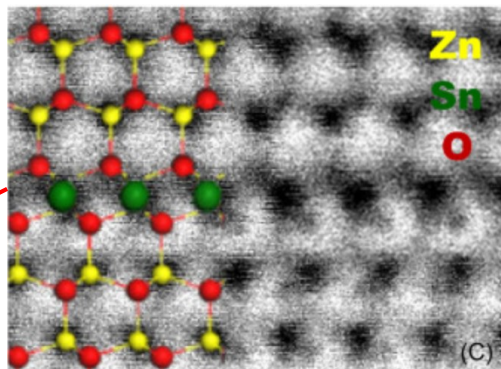
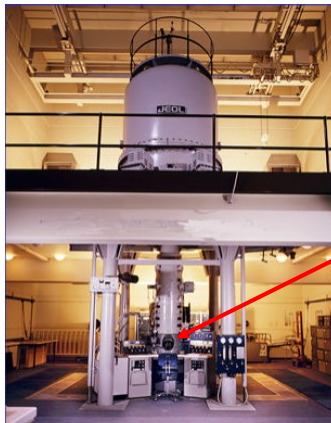


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Microstructure

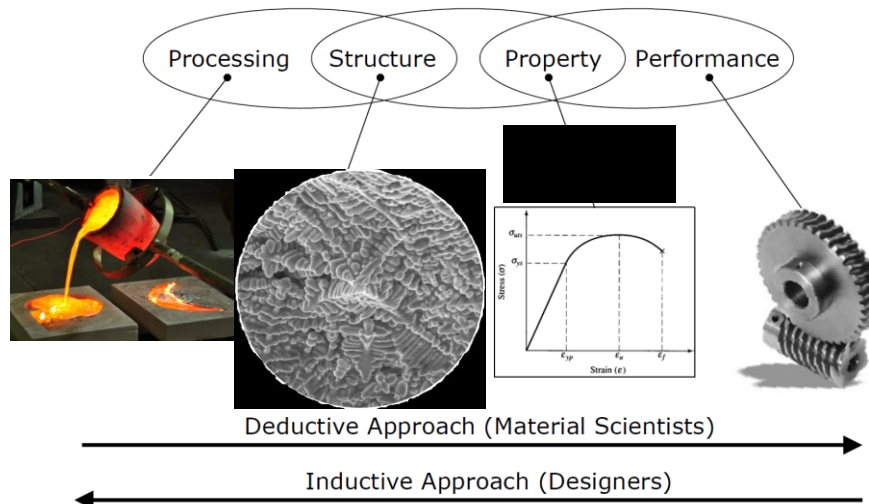


Microstructure Observation: Transmission Electron Microscope



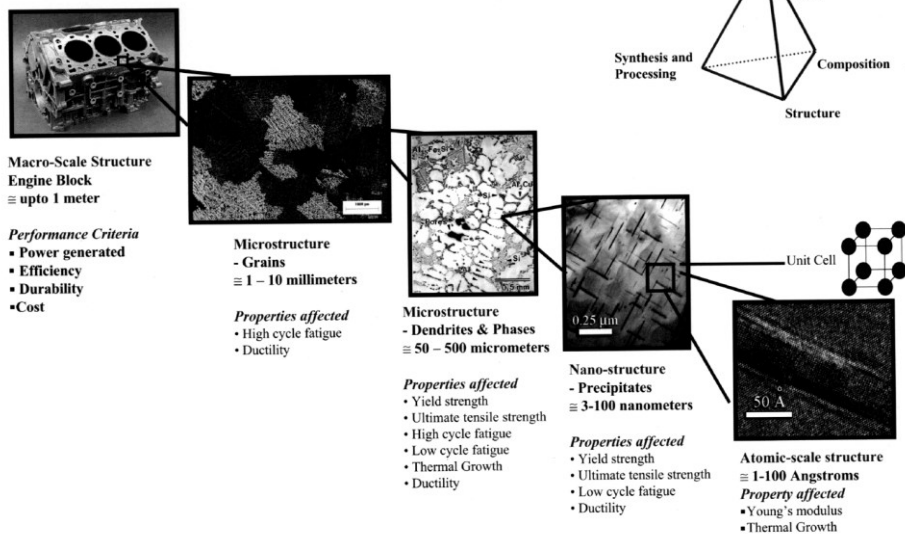
1250 keV U of Antwerp (Belgium)

Property, Processing, Performance and Structure Interactions



Olson, G. B., 1997, "Computational Design of Hierarchically Structured Materials," Science, Vol. 277, No. 5330, pp. 1237-1242.

Microstructure Multiscale



A real-world example of important microstructural features at different length-scales, resulting from the sophisticated synthesis and processing used, and the properties they influence. The atomic, nano, micro, and macro-scale structures of cast aluminum alloys (for engine blocks) in relation to the properties affected and performance are shown. The materials science and engineering (MSE) tetrahedron that represents this approach is shown in the upper right corner.

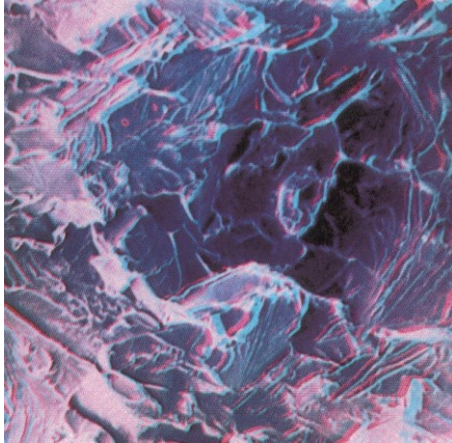
(Illustrations Courtesy of John Allison and William Donlon, Ford Motor Company).



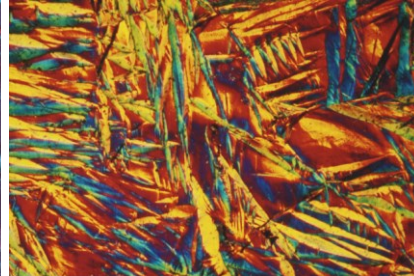
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Strereological Microstructures

Normasky Interference Contrast



fracture surface



Martensite: BCT matrix (brown) + Iron carbide (green)



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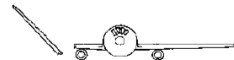
Performances versus Properties



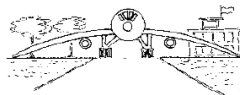
← Insufficiently **stiff**
(require: $> E$) **Young modulus**



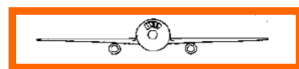
← Insufficiently **strong**
(require: $> \sigma_y$) **Yield strength**



← Insufficient **tough**
(require: $> K_{IC}$) **Fracture toughness**



← Too **heavy**
(require: $< \rho$) **Density**



← { **stiff**
strong
tough
light } ← **Correct selection
of Materials
+
Adequate
Component design**



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Assignment n.1



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Remind from STM

Property	Ionic Bond	Covalent Bond	Metallic Bond
Hardness	High	Low → Very High	Low → High
Ductility	Brittle	Brittle	Ductile
Melting Temperature	High	Low → Very High	Low → High
Electrical and Thermal Conductivity	Low	Low	High

Task of the Assignment n.1

- Review and learn the individual properties of the **6 big metals** listed hereinafter along all listed properties in the table;
- As these metals lead to **6 major classes of inherent alloys** of major engineering relevance, select two them and discuss the two alloys formed assuming one element as main element of the alloy and the other as alloying element and vice versa, then describe in the two alloys the:
 - main properties imparted by the primary element
 - Main strengthening methods (mechanism to make the alloy stronger)
 - role played by the main alloy-element in terms of high strength and toughness
 - main engineering applications and properties required for its performance
 - Individual sustainability factors (as compared to reference steels), in terms of energy consumption, CO₂ emission, water and primary materials consumption during manufacturing and recycling rate



The big 3

	Iron	Aluminium	Copper
Available?			
Cheap ? (£ / tonne 1997)			
M.P. (°C)			
Density (kg m ⁻³)			
Allotropy			
Dissolves things ?			
T-Variable solubility			
Compounds			
Stability			
Hazardous ?			



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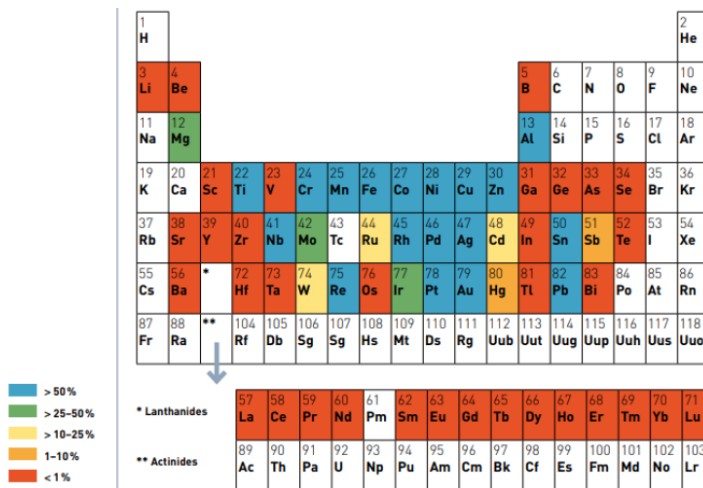
The «other» 3

	Nickel	Titanium	Magnesium
Available?			
Cheap ? (£ / tonne 1997)			
M.P. (°C)			
Density (kg m ⁻³)			
Allotropy			
Dissolves things ?			
T-Variable solubility			
Compounds			
Stability			
Hazardous ?			



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Recycling Rate



END of Lecture 1

Self-study
Historical Metals Curiosities
hereinafter